1. What is an Operating System? Explain any four functions with examples.

Definition:

An **Operating System (OS)** is system software that acts as an interface between the user and computer hardware. It manages hardware resources, provides services to applications, and ensures efficient execution of programs.

Four Functions of an OS:

1. **Process Management**
    - Handles creation, scheduling, and termination of processes.
    - Example: In Windows, Task Manager shows running processes and allows you to end them.
2. **Memory Management**
    - Allocates and deallocates memory space to processes.
    - Example: When you open multiple tabs in Chrome, the OS ensures each tab gets separate memory space.
3. **File System Management**
    - Manages files on storage devices, including creation, deletion, and access control.
    - Example: Linux uses the ext4 file system to organize files in directories.
4. **Device Management**
    - Controls hardware devices through drivers, ensuring smooth input/output operations.
    - Example: Printing a document — the OS sends data to the printer via its driver.

(Other functions include security, networking, and user interface management.)

2. Differentiate between BIOS and UEFI. Also write the full forms of MBR and GPT.

BIOS (Basic Input/Output System):

- Legacy firmware used in older computers.
- Runs in 16-bit mode, limited to 1 MB of addressable space.
- Uses MBR (Master Boot Record) for booting.
- Slower boot process.

- Example: Old PCs with Windows XP used BIOS.

UEFI (Unified Extensible Firmware Interface):

- Modern replacement for BIOS.
- Runs in 32-bit or 64-bit mode, supports larger storage.
- Uses GPT (GUID Partition Table) for booting.
- Faster booting, supports secure boot.
- Example: Modern laptops with Windows 10/11 use UEFI.

Full Forms:

- **MBR:** Master Boot Record
- **GPT:** GUID Partition Table

3. Explain the difference between User Mode and Kernel Mode. What is the role of the Kernel in an Operating System?

User Mode:

- Restricted mode where applications run.
- Cannot directly access hardware or memory.
- Example: Running MS Word or a browser — they operate in user mode.

Kernel Mode:

- Privileged mode where the OS kernel runs.
- Has full access to hardware and system resources.
- Example: Device drivers and system calls execute in kernel mode.

Role of the Kernel:

- Core component of the OS, responsible for:
    A. **Process Management** – scheduling and execution.
    B. **Memory Management** – allocation and protection.
    C. **Device Management** – communication with hardware via drivers.
    D. **System Calls Handling** – provides interface between user applications and hardware.